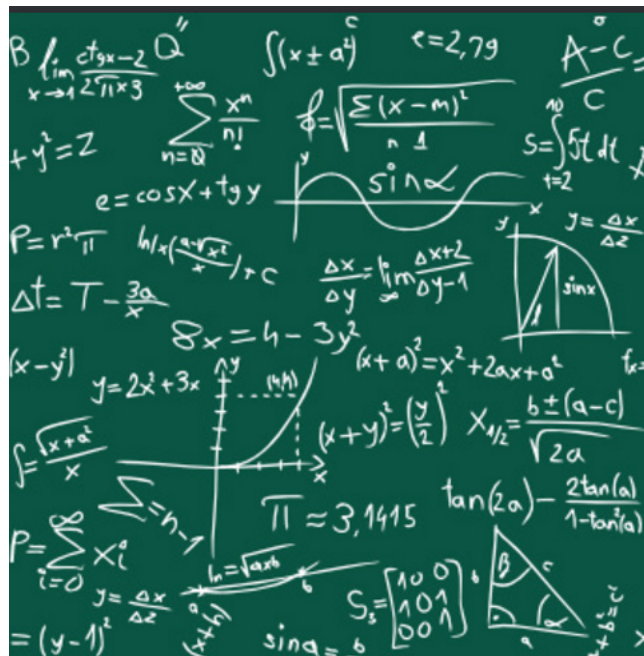


Factorials, binomial coefficients and triangle inequality



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- **Book:** *“Analisi matematica 1”*, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Factorials

Definition 1: factorial of n

The factorial of n is the product of the first n positive integers. It is denoted by $n!$ and is read “ n factorial”. In formulas:

$$n! = \prod_{k=1}^n k = 1 \cdot 2 \cdot 3 \cdot \dots \cdot (n-1) \cdot n$$

By definition, we set $0! = 1$.

- The number $n!$ grows very rapidly as n increases. The first values are:

n	0	1	2	3	4	5	6	7	8	9	10
$n!$	1	1	2	6	24	120	720	5.040	40.320	362.880	3.628.800

- Some properties of the factorial, which are immediate to verify, are:

$$n! = n \cdot (n-1)!$$

$$\frac{n!}{(n-k)!} = n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot (n-k+1), \text{ with } k \geq 1 \quad (1)$$

With $k \geq 1$, it becomes the product of k factors, starting from n and decreasing by one unit at a time.

Example 1: Computing the factorial

$$\frac{100!}{95!} = \frac{100!}{(100-5)!} = 100 \cdot 99 \cdot 98 \cdot 97 \cdot 96 = 9.034.502.400$$

It is always convenient to simplify expressions containing the factorial as much as possible, before computing them!

2 Binomial coefficients

Definition 2: binomial coefficient

The **binomial coefficient** is defined as the number:

$$c_{n,k} = \frac{n!}{k!(n-k)!} \quad \text{with } 0 \leq k \leq n. \quad (2)$$

The binomial coefficient $c_{n,k}$ is usually denoted by the symbol: $\binom{n}{k}$ which is read “ n choose k ”.

- Given rule (1), we have:

$$c_{n,k} = \frac{n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot (n-k+1)}{k!} \quad (3)$$

with $k \geq 1$, an expression that is more convenient for computing the binomial coefficient.

We have:

$$\binom{n}{n-k} = \binom{n}{k} \quad (4)$$

Since:

$$\binom{n}{n-k} = \frac{n!}{(n-k)!(n-(n-k))!} = \frac{n!}{k!(n-k)!} = \binom{n}{k}$$

We have:

$$\binom{n-1}{k-1} + \binom{n-1}{k} = \binom{n}{k} \quad (5)$$

Since:

$$\begin{aligned} \binom{n-1}{k-1} + \binom{n-1}{k} &= \frac{(n-1)!}{(k-1)! \underbrace{(n-1-(k-1))!}_{=(n-k)! = (n-k)(n-k-1)!}} + \frac{(n-1)!}{k!(n-k-1)!} \\ &= \frac{(n-1)!}{(k-1)!(n-k)(n-k-1)!} + \frac{(n-1)!}{k(k-1)!(n-k-1)!} \\ &= \frac{k(n-1)! + (n-k)(n-1)!}{k(k-1)!(n-k)(n-k-1)!} = \frac{\overbrace{(n-1)!}^{=n!} n}{\underbrace{k(k-1)!}_{k!} \underbrace{(n-k)(n-k-1)!}_{(n-k)!}} \\ &= \frac{n!}{k!(n-k)!} = \binom{n}{k} \end{aligned}$$

It also follows that:

$$\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}$$

2.1 Newton's formula

The n -th power of a binomial $(a + b)$ can be computed with the following formula (from which the name binomial coefficient derives):

Remark 1: Newton's formula

For every integer $n \geq 0$, with $a, b \in \mathbb{R}$, we have:

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k \quad (6)$$

Proof. By induction on n .

- **Base case of the induction**

Let $n = 0$. Then the statement becomes: $(a + b)^0 = \binom{0}{0} a^0 b^0$ i.e. $1 = 1$ which is clearly true.

- **Inductive step**

Suppose it is true for n , and let us prove it for $(n + 1)$. By the inductive hypothesis, we have: $(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$. Then:

$$\begin{aligned}
 (a + b)^{n+1} &= (a + b) \cdot (a + b)^n = (a + b) \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k \\
 &= \sum_{k=0}^n \binom{n}{k} a^{n-k+1} b^k + \underbrace{\sum_{k=0}^n \binom{n}{k} a^{n-k} b^{k+1}}_{= \sum_{k=1}^{n+1} \binom{n}{k-1} a^{n-k+1} b^k} \\
 &= a^{n+1} + \sum_{k=1}^n \binom{n}{k} a^{n-k+1} b^k + b^{n+1} + \sum_{k=1}^n \binom{n}{k-1} a^{n-k+1} b^k \\
 &= a^{n+1} + b^{n+1} + \sum_{k=1}^n \underbrace{\left(\binom{n}{k} + \binom{n}{k-1} \right)}_{= \binom{n+1}{k}} a^{n-k+1} b^k \\
 &= \sum_{k=0}^{n+1} \binom{n+1}{k} a^{n+1-k} b^k
 \end{aligned}$$

which is exactly the desired statement, for $n + 1$.

□

Remark 2

For every integer $n \geq 0$ and integer k such that $0 \leq k \leq n$, we have:

$$\sum_{k=0}^n \binom{n}{k} = 2^n \tag{7}$$

Proof. We write:

$$2^n = (1 + 1)^n$$

We now apply Newton's formula (6):

$$(1 + 1)^n = \sum_{k=0}^n \binom{n}{k} 1^{n-k} 1^k = \sum_{k=0}^n \binom{n}{k}$$

□

2.2 Recursive computation of binomial coefficients

Relation (5) allows us to compute the binomial coefficients $\binom{n}{k}$ by means of the so-called **Pascal's triangle** (or Tartaglia's triangle).

The rules for building the triangle are:

1. At the top of the triangle we place the number $\binom{0}{0} = 1$ (by definition).
2. On the sides we place the numbers $\binom{n}{0} = \binom{n}{n} = 1$ for every $n \geq 1$.
3. For $0 < k < n$, the number $\binom{n}{k}$ is written at the intersection of the n -th row and the k -th column.
4. The number $\binom{n}{k}$ is the sum of the two numbers located in the previous row, the one in the same column and the one in the previous column.

Example 2: Pascal's triangle (or Tartaglia's triangle)

Pascal's triangle with $n \leq 10$ and $k \leq 10$ is:

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$	$k=10$
$n=0$	1										
$n=1$	1	1									
$n=2$	1	2	1								
$n=3$	1	3	3	1							
$n=4$	1	4	6	4	1						
$n=5$	1	5	10	10	5	1					
$n=6$	1	6	15	20	15	6	1				
$n=7$	1	7	21	35	35	21	7	1			
$n=8$	1	8	28	56	70	56	28	8	1		
$n=9$	1	9	36	84	126	126	84	36	9	1	
$n=10$	1	10	45	120	210	252	210	120	45	10	1

To compute the binomial coefficient $\binom{5}{3}$, corresponding to the blue cell, we can use relation (5) and add the two binomial coefficients $\binom{4}{2}$ and $\binom{4}{3}$, corresponding to the red cells:

$$\binom{5}{3} = \binom{4}{2} + \binom{4}{3} = 6 + 4 = 10.$$

Example 3: computing the power of a binomial using Pascal's triangle

Thanks to the previous Pascal's triangle and to Newton's formula (6) we can compute:

$$(a+b)^5 = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5$$

3 Absolute value

Definition 3: absolute value

The **absolute value** of a real number $a \in \mathbb{R}$ (or **modulus** of a) is the non-negative number defined as follows:

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases} \quad (8)$$

From the definition of absolute value it immediately follows that:

$$\forall \varepsilon \geq 0, a \in \mathbb{R}, \quad |a| \leq \varepsilon \iff -\varepsilon \leq a \leq \varepsilon \quad (9)$$

3.1 Triangle inequality in $\mathbb{R}$

Remark 3: triangle inequality in $\mathbb{R}$

$$|b + c| \leq |b| + |c| \quad \forall b, c \in \mathbb{R} \quad (10)$$

Proof. We write the two relations:

$$-|b| \leq b \leq |b|, \quad -|c| \leq c \leq |c|$$

and add them side by side:

$$-(|b| + |c|) \leq b + c \leq |b| + |c|$$

Hence, by (9), with $\varepsilon = |b| + |c| \geq 0$ and $a = b + c$, (10) follows. $\square$

- The triangle inequality is also used in the following form:

$$|d - e| \leq |d - f| + |e - f| \quad \forall d, e, f \in \mathbb{R} \quad (11)$$

To obtain it, it suffices to set in (10):

$$b = d - f, \quad c = f - e$$

we obtain:

$$|d - f + f - e| = |d - e| \leq |d - f| + |f - e| = |d - f| + |e - f|$$

since:

$$|f - e| = |e - f| \quad \forall e, f \in \mathbb{R}$$

- Moreover, the triangle inequality can also be written in the following form:

$$|g| \leq |g - h| + |h| \quad \text{i.e.} \quad |g| - |h| \leq |g - h| \quad \forall g, h \in \mathbb{R} \quad (12)$$

To obtain it, it suffices to set in (10):

$$b = g - h, \quad c = h$$

Remark 4: reverse triangle inequality in $\mathbb{R}$

$$||g| - |h|| \leq |g - h|, \quad \forall g, h \in \mathbb{R}. \quad (13)$$

Proof. From (12) we have

$$|g| - |h| \leq |g - h| \quad \forall g, h \in \mathbb{R}$$

Similarly, swapping g and h in (12) we obtain:

$$|h| - |g| \leq |h - g| = |g - h| \quad \text{that is} \quad |g| - |h| \geq -|g - h|$$

hence we have

$$-(|g - h|) \leq |g| - |h| \leq |g - h| \quad \forall g, h \in \mathbb{R}$$

Hence, by (9), with $\varepsilon = |g - h| \geq 0$ and $a = |g| - |h|$, the reverse triangle inequality follows. $\square$

- Inequality (10) can easily be extended to the case of k terms:

$$\left| \sum_{i=1}^k b_i \right| \leq \sum_{i=1}^k |b_i|. \quad (14)$$

- The following immediate properties also hold:

$$|bc| = |b||c|, \quad \left| \frac{b}{c} \right| = \frac{|b|}{|c|}, \quad |-b| = |b| \quad \forall b, c \in \mathbb{R}. \quad (15)$$